When to Surf the Atlantic Coast?

Surfable swell in Hossegor, Peniche and Taghazout

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MAT2007 Introduction to Programming

15/10/2026

Introduction

To make the most of a surfing trip, certain things matter. If the swell is too small, there is nothing to ride, and if it is too large, the sea becomes unsurfable. Surfers therefore plan their trips based on where and when to go, and that is exactly what this report investigates for three popular surfing destinations: **in which months is the swell most surfable in Hossegor (France), Peniche (Portugal), and Taghazout (Morocco), and how consistent is that from year to year?**

Dataset

To answer this question, hourly values of wave height H and wave period T for these regions were retrieved from the ERA5-Ocean reanalysis (Hersbach et al., 2020), accessed through the Open-Meteo Marine API (Open-Meteo, n.d.). The data cover 2016–2025 (10 full years), with 87 672 hourly values per region and no missing values. ERA5-Ocean has a grid spacing of 0.5°, which is about 50 km, so each region is represented by the nearest offshore grid point (Hossegor 43.5°N 2.0°W, Peniche 39.5°N 10.0°W, Taghazout 30.5°N 10.0°W). The data therefore describe the offshore swell climate and not the surf at a specific break.

Method

Hourly values were averaged to daily means. A day is called surfable if $1\,\mathrm{m} \leq H \leq 3\,\mathrm{m}$ and $T \geq 8$ s. For each region, month and year, the fraction f_y of surfable days was computed; each year is one subsample. The result for a month is the mean over $N = 10$ years with the standard error

$$\bar{f} = \frac{1}{N}\sum_{y=1}^{N} f_y, \qquad \sigma_{\bar{f}} = \frac{s}{\sqrt{N}}, \tag{1}$$

where s is the standard deviation of the f_y. To explain Peniche's flat winter curve, the mean wave height and the share of days with $H > 3$ m were also computed. Two values are compared using $\sigma_\Delta = \sqrt{\sigma_1^2 + \sigma_2^2}$. Winter is defined as December to February, the core of the swell season (Fig. 1, left); the value of a year uses these three months of that calendar year. The analysis was done in Python (pandas, matplotlib).

Results and discussion

Figure 1 (left) shows the share of surfable days for each month. All three regions follow the same seasonal pattern, with the fewest surfable days in July (Table 1). Taghazout reaches the highest value, 79.7 ± 3.0 % in January, but January is hardly better than December or February ($0.5\,\sigma_\Delta$ and $0.8\,\sigma_\Delta$), so it is fairer to speak of a winter plateau of 77–80 %. This is clearly higher than

Peniche's best month, October, by 17.8±5.7 percentage points ($3.1\,\sigma_\Delta$). Hossegor peaks in February ($71.5 \pm 3.7\,\%$), which is not significantly different from Taghazout in that month ($76.7 \pm 2.1\,\%$). Peniche stays lower and flatter.

The second part of the question is how consistent this is from year to year (Fig. 1, right). Taghazout had more surfable winter days than both other regions in all 10 winters. Its winter value ranged from 73.5 to 82.2 % (mean 78.0 %, $s = 2.5$ points), compared with 51.4–71.5 % for Hossegor ($s = 5.1$) and 38.2–66.1 % for Peniche ($s = 9.3$). So Taghazout is not only the best but also the most predictable region, and October to March gives the same ranking.

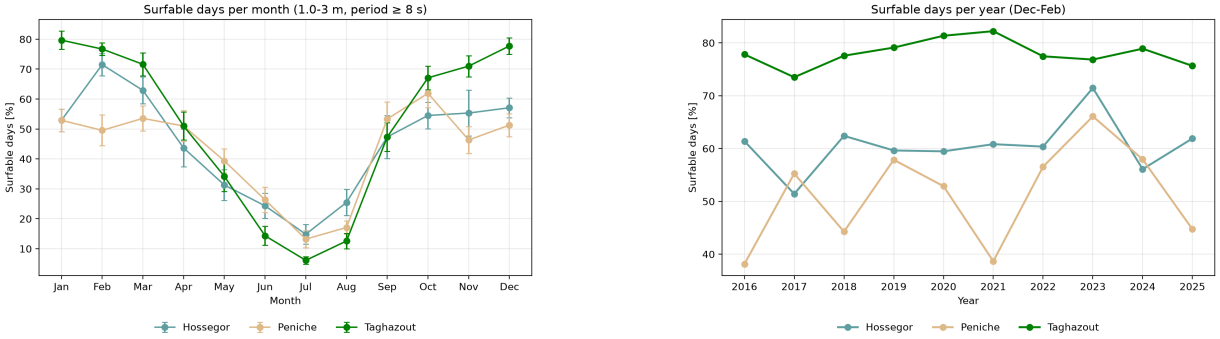


Figure 1: **Left:** percentage of surfable days (1–3 m, $T \geq 8$ s) per month, averaged over 2016–2025; error bars are the standard error over the 10 years. **Right:** percentage of surfable days in winter (Dec–Feb), per year.

Table 1: Peak and minimum of the surfable-day fraction, and the share of winter (Dec–Feb) days with a daily mean $H > 3$ m.

Region	Peak month	July [%]	Winter days $H > 3$ m [%]
Hossegor	Feb: 71.5 ± 3.7	14.8 ± 3.3	≈ 17
Peniche	Oct: 61.9 ± 4.8	13.2 ± 2.9	≈ 41
Taghazout	Jan: 79.7 ± 3.0	6.1 ± 1.2	≈ 13

Peniche's flat winter curve has a simple reason: its swell is often too big, not too small. Its mean wave height is the highest of the three regions in every month from October to April (February: 3.07 ± 0.13 m against 2.29 ± 0.09 m at Taghazout), and about 41 % of its winter days exceed 3 m, compared with 13–17 % elsewhere. With an upper limit of 2.5 m instead, all curves drop by 10–20 points, but Taghazout still leads in December and January.

Conclusion

Taghazout shows the most reliable surfable swell, with about 78 % of days surfable from December to February and the best result in all 10 winters. Hossegor follows, peaking at 71 % in February, while Peniche has less surfable days as its swell is often too big. Limitations are the simple thresholds, which depend on the surfer's level, the 50 km grid spacing, and the missing wind and tide data.

References

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